

Electrical and Electronics Engineering Basics

A focused undergraduate reference: circuits, electronics, machines, digital, controls, measurement, and power

Study Guide

July 23, 2026

Abstract

This reference collects the models and equations used repeatedly in introductory electrical and electronics engineering. It begins with DC networks and transients, develops sinusoidal and three-phase analysis, then covers semiconductor devices, op-amp circuits, transformers and machines, digital electronics, feedback control, measurement, safety, and introductory power systems. Every worked result uses the passive sign convention unless stated otherwise: current entering a positive-labeled terminal means the element absorbs positive power.

How to use this guide

For any circuit: (1) choose voltage polarities and current directions, (2) state the operating condition—DC steady state, switching transient, or sinusoidal steady state—and (3) choose a model appropriate to that condition. A negative answer means the actual direction is opposite the assumed reference direction.

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1 Core Electrical Quantities and DC Networks

1.1 Quantities, signs, and power

Voltage is energy per unit charge, current is charge flow, and resistance opposes current:

$$i = \frac{dq}{dt}, \quad v = \frac{dw}{dq}, \quad v = Ri, \quad p(t) = v(t)i(t), \quad w = \int p(t) dt. \quad (1)$$

The SI units are volt (V), ampere (A), ohm (Ω), watt (W), joule (J), farad (F), henry (H), and hertz (Hz). Power is absorbed when $p > 0$ and delivered when $p < 0$ under the passive sign convention.

1.2 KCL, KVL, and basic combinations

Kirchhoff's current law (KCL) expresses charge conservation at a node:

$$\sum i_{\text{entering}} = \sum i_{\text{leaving}}. \quad (2)$$

Kirchhoff's voltage law (KVL) expresses energy conservation around a closed loop:

$$\sum v_{\text{rises}} = \sum v_{\text{drops}}. \quad (3)$$

For resistors, series elements share current and parallel elements share voltage:

$$R_{\text{series}} = \sum_k R_k, \quad \frac{1}{R_{\text{parallel}}} = \sum_k \frac{1}{R_k}, \quad G = \frac{1}{R}. \quad (4)$$

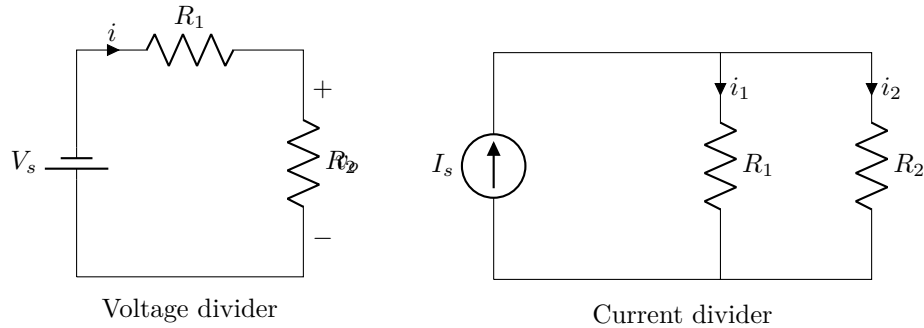


Figure 1: Divider circuits and reference directions.

Network	Useful result	Condition
Voltage divider	$v_o = V_s \frac{R_2}{R_1 + R_2}$	Unloaded output
Current divider, two branches	$i_1 = I_s \frac{R_2}{R_1 + R_2}$	$R_1 \parallel R_2$
Power in a resistor	$P = VI = I^2 R = V^2/R$	Passive sign convention

1.3 Network methods

Nodal analysis solves unknown node voltages with KCL; it is efficient for current sources and parallel branches. **Mesh analysis** solves loop currents with KVL; it is efficient for voltage sources and planar networks. **Superposition** finds a linear-circuit response one independent source at a time: replace other independent voltage sources by shorts and independent current sources by opens. Dependent sources remain active. Do not superpose power because it is nonlinear.

1.4 Source transformation, Thevenin, Norton, and maximum power

A voltage source V_s in series with R_s is equivalent at its terminals to a current source $I_s = V_s/R_s$ in parallel with R_s . Any linear two-terminal network can be reduced to:

$$\text{Thevenin: } V_{\text{th}} \text{ in series with } R_{\text{th}}, \quad \text{Norton: } I_N \text{ in parallel with } R_N, \quad (5)$$

where $R_N = R_{\text{th}}$, $V_{\text{th}} = I_N R_{\text{th}}$, and V_{th} is the open-circuit voltage. Find R_{th} by deactivating independent sources or by $R_{\text{th}} = V_{\text{test}}/I_{\text{test}}$ when dependent sources exist. A resistive load receives maximum power at $R_L = R_{\text{th}}$, with $P_{L,\text{max}} = V_{\text{th}}^2/(4R_{\text{th}})$.

Worked example: Loaded voltage divider

A 12 V source feeds $R_1 = 2\text{ k}\Omega$, with $R_2 = 4\text{ k}\Omega$ to ground. A $4\text{ k}\Omega$ load is connected across R_2 . The load is parallel with R_2 , so $R_2 \parallel R_L = 2\text{ k}\Omega$. Therefore

$$v_o = 12\text{ V} \frac{2}{2+2} = 6\text{ V}.$$

The unloaded result would be 8 V; a real load changes a divider unless its resistance is much larger than the lower divider resistance.

2 Capacitors, Inductors, and First-Order Transients

2.1 Element laws and DC behavior

$$i_C = C \frac{dv_C}{dt}, \quad w_C = \frac{1}{2} C v_C^2, \quad Z_C = \frac{1}{j\omega C}, \quad (6)$$

$$v_L = L \frac{di_L}{dt}, \quad w_L = \frac{1}{2} L i_L^2, \quad Z_L = j\omega L. \quad (7)$$

A capacitor voltage cannot jump without an impulse current: $v_C(0^+) = v_C(0^-)$. An inductor current cannot jump without an impulse voltage: $i_L(0^+) = i_L(0^-)$. In a DC steady state, an ideal capacitor is an open circuit and an ideal inductor is a short circuit. At high frequency, $|Z_C|$ decreases and $|Z_L|$ increases.

	RC circuit	RL circuit
State variable	Capacitor voltage v_C	Inductor current i_L
Continuity	$v_C(0^+) = v_C(0^-)$	$i_L(0^+) = i_L(0^-)$
Time constant	$\tau = R_{\text{eq}} C$	$\tau = L/R_{\text{eq}}$
DC steady state	Capacitor open circuit	Inductor short circuit
Natural response	$v_C(t) = V_0 e^{-t/RC}$	$i_L(t) = I_0 e^{-tR/L}$

2.2 Universal first-order response

For a switch at $t = 0$, first solve the circuit before switching for the initial condition, then solve the circuit after switching at $t \rightarrow \infty$ for the final condition. Any first-order state variable has the form

$$x(t) = x(\infty) + [x(0^+) - x(\infty)] e^{-t/\tau}, \quad t \geq 0. \quad (8)$$

After 5τ , the response is within about 0.7% of its final value. A **source-free** circuit has no independent sources after switching, so its stored energy decays exponentially. A step response includes a nonzero final value.

Worked example: RC charging and RL current rise

For $V_s = 10\text{ V}$, $R = 2\text{ k}\Omega$, and $C = 50\text{ }\mu\text{F}$, $\tau = RC = 0.10\text{ s}$. An initially uncharged capacitor charges as

$$v_C(t) = 10(1 - e^{-t/0.10})\text{ V}.$$

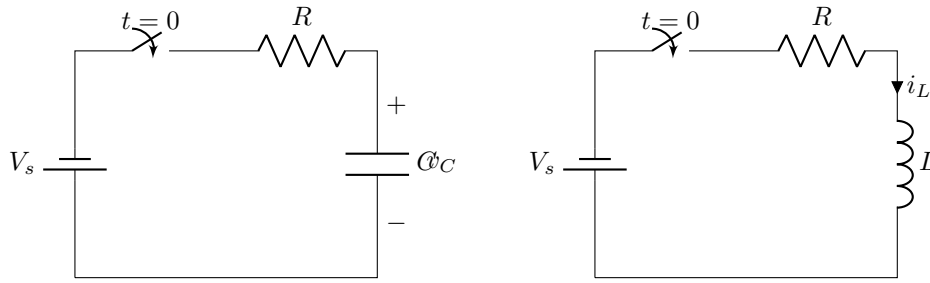


Figure 2: RC and RL step-response circuits.

At $t = 0.10$ s, $v_C = 6.32$ V. If instead $L = 0.20$ H and $R = 10\ \Omega$, then $\tau = L/R = 0.020$ s and an initially zero inductor current rises to $i_L(t) = (10/10)(1 - e^{-t/0.020})$ A.

3 Sinusoidal AC, Phasors, RLC, and Filters

3.1 Sinusoids, RMS, and phasors

For $v(t) = V_m \cos(\omega t + \theta)$, $f = \omega/(2\pi)$ and $T = 1/f$. A sinusoid has $V_{\text{rms}} = V_m/\sqrt{2}$ and zero average over a full cycle. The average of a full-wave rectified sine is $2V_m/\pi$. A phasor uses an RMS magnitude by convention:

$$v(t) = \sqrt{2} \operatorname{Re}\{\underline{V}e^{j\omega t}\}, \quad \underline{V} = V_{\text{rms}}\angle\theta. \quad (9)$$

In phasor form, $d/dt \rightarrow j\omega$ and KCL/KVL still apply.

3.2 Impedance and phase

Element	Impedance Z	Current versus voltage	Average real power	DC limit
Resistor	R	In phase	$I_{\text{rms}}^2 R$	R
Inductor	$j\omega L$	Current lags 90°	0 ideal	Short
Capacitor	$1/(j\omega C)$	Current leads 90°	0 ideal	Open

Admittance is $Y = 1/Z = G + jB$. In a series RLC circuit,

$$Z = R + j\left(\omega L - \frac{1}{\omega C}\right), \quad I = V/Z. \quad (10)$$

Resonance occurs at $\omega_0 = 1/\sqrt{LC}$, where the series impedance is minimum and current is maximum. For a series RLC, $Q = \omega_0 L/R$ and the approximate bandwidth is $\Delta\omega = \omega_0/Q = R/L$.

3.3 Power in AC circuits

Let $\phi = \theta_v - \theta_i$. Instantaneous power is $p(t) = v(t)i(t)$; its average over a cycle is real power:

$$P = V_{\text{rms}} I_{\text{rms}} \cos \phi, \quad Q = V_{\text{rms}} I_{\text{rms}} \sin \phi, \quad \underline{S} = P + jQ = \underline{V}\underline{I}^*, \quad |\underline{S}| = V_{\text{rms}} I_{\text{rms}}. \quad (11)$$

The power factor is $\text{pf} = P/|\underline{S}| = \cos \phi$. Inductive loads are lagging ($Q > 0$) and capacitive loads are leading ($Q < 0$).

Worked example: Series RL power

A 120 V_{rms}, 60 Hz supply feeds $R = 20\ \Omega$ and $L = 53.1$ mH. Then $X_L = \omega L \approx 20\ \Omega$, $Z = 20 + j20 = 28.3\angle 45^\circ\ \Omega$, and $I = 4.24\angle -45^\circ$ A. Thus $P = I^2 R = 360$ W, $Q = I^2 X_L = 360$ var, and $\text{pf} = 0.707$ lagging.

3.4 Transfer functions and first-order filters

The transfer function is $H(s) = Y(s)/X(s)$ with zero initial conditions. Replace d/dt by s to obtain it from circuit equations; frequency response is $H(j\omega)$. For a resistor-capacitor low-pass with output across C :

$$H_{LP}(s) = \frac{1}{1 + sRC}, \quad f_c = \frac{1}{2\pi RC}, \quad |H_{LP}(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}. \quad (12)$$

For the high-pass output across R :

$$H_{HP}(s) = \frac{sRC}{1 + sRC}. \quad (13)$$

At f_c , magnitude is $1/\sqrt{2}$ (-3 dB). A first-order magnitude slope is 20 dB/decade beyond its corner: down for low-pass, up for high-pass.

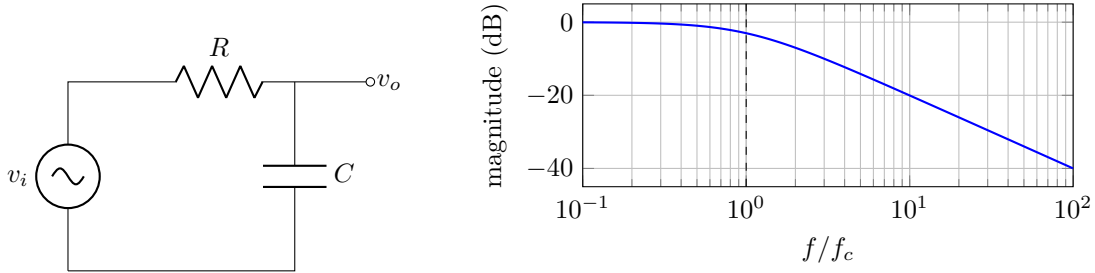


Figure 3: RC low-pass filter and its exact magnitude response.

4 Balanced Three-Phase Circuits and Y-Delta Networks

4.1 Three-phase basics

Balanced phase voltages have equal magnitudes separated by 120° . Positive sequence is $a \rightarrow b \rightarrow c$. A balanced load may be analyzed one phase at a time. Use RMS line-to-line voltage V_{LL} and line current I_L for total three-phase power:

$$\underline{S}_{3\phi} = \sqrt{3} \underline{V}_{LL} \underline{I}_L^*, \quad P = \sqrt{3} V_{LL} I_L \cos \phi, \quad Q = \sqrt{3} V_{LL} I_L \sin \phi. \quad (14)$$

Relationship	Balanced Y (wye/star) load	Balanced Delta load
Connection	Each impedance phase-to-neutral	Each impedance line-to-line
Phase voltage	$V_\phi = V_{LL}/\sqrt{3}$	$V_\phi = V_{LL}$
Line/phase current	$I_L = I_\phi$	$I_L = \sqrt{3} I_\phi$
Phase current	$I_\phi = V_\phi / Z_\phi$	$I_\phi = V_\phi / Z_\phi$
Per-phase power	$S_\phi = V_\phi I_\phi^*$	$S_\phi = V_\phi I_\phi^*$
Total balanced power	$S_{3\phi} = 3S_\phi = \sqrt{3} V_{LL} I_L^*$	Same result
Equal line-current impedance	$Z_Y = Z_\Delta / 3$	$Z_\Delta = 3Z_Y$

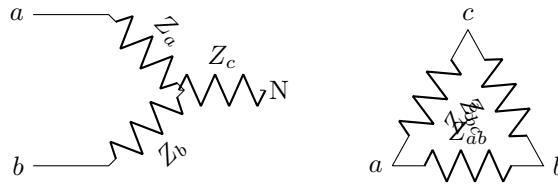


Figure 4: Wye and delta impedance networks.

4.2 Delta-ye transformation

For delta impedances Z_{ab} , Z_{bc} , Z_{ca} and $\Sigma_\Delta = Z_{ab} + Z_{bc} + Z_{ca}$:

$$Z_a = \frac{Z_{ab}Z_{ca}}{\Sigma_\Delta}, \quad Z_b = \frac{Z_{ab}Z_{bc}}{\Sigma_\Delta}, \quad Z_c = \frac{Z_{bc}Z_{ca}}{\Sigma_\Delta}. \quad (15)$$

For wye impedances Z_a , Z_b , Z_c and $\Sigma_Y = Z_aZ_b + Z_bZ_c + Z_cZ_a$:

$$Z_{ab} = \frac{\Sigma_Y}{Z_c}, \quad Z_{bc} = \frac{\Sigma_Y}{Z_a}, \quad Z_{ca} = \frac{\Sigma_Y}{Z_b}. \quad (16)$$

Worked example: Balanced delta load power

Three equal $12\angle 30^\circ \Omega$ impedances form a delta load on a 208 V line-to-line source. Each branch sees 208 V, so $I_\phi = 208/12 = 17.33$ A. The line current magnitude is $I_L = \sqrt{3}(17.33) = 30.0$ A. With $\text{pf} = \cos 30^\circ = 0.866$ lagging,

$$P = \sqrt{3}(208)(30)(0.866) = 9.36 \text{ kW}, \quad Q = 5.40 \text{ kvar}.$$

4.3 Power-factor correction

To improve a lagging load from ϕ_1 to ϕ_2 while retaining real power P , add a shunt capacitor supplying

$$Q_C = P(\tan \phi_1 - \tan \phi_2), \quad C = \frac{Q_C}{\omega V^2}. \quad (17)$$

Use phase voltage V for a per-phase capacitor or line-to-line voltage for a delta-connected capacitor bank. Capacitors reduce source current but do not reduce the load's real-power demand.

5 Diodes, BJTs, and MOSFETs

5.1 PN junctions and diodes

A diode conducts strongly in forward bias and blocks current in reverse bias until breakdown. Common circuit models are ideal ($V_D = 0$ when on), constant drop ($V_D \approx 0.7$ V for silicon), and exponential:

$$i_D = I_S \left(e^{v_D/(nV_T)} - 1 \right), \quad V_T \approx 25.9 \text{ mV at } 300 \text{ K}. \quad (18)$$

A Zener diode is designed for reverse breakdown regulation. A capacitor-filtered full-wave rectifier has ripple approximately $\Delta V \approx I_{load}/(2fC)$; use only after confirming capacitor voltage and ripple-current ratings.

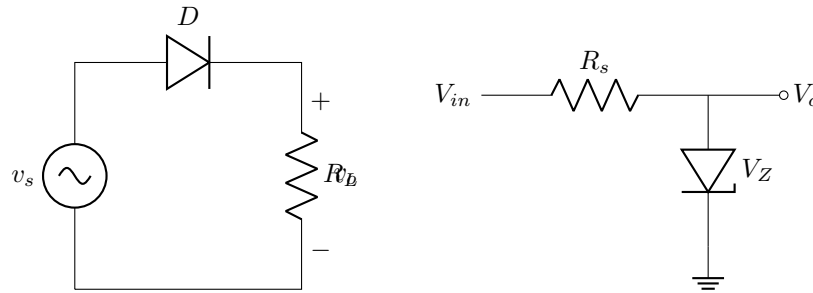


Figure 5: Half-wave rectifier and Zener shunt regulator.

5.2 BJT and MOSFET essentials

An NPN BJT operates in cutoff (off), forward active (amplification), and saturation (switch on). In forward active operation,

$$I_C \approx \beta I_B, \quad I_E = I_C + I_B, \quad V_{BE} \approx 0.7 \text{ V}. \quad (19)$$

For switching, choose sufficient base current to force saturation and respect collector current, V_{CE} , power, and switching-speed limits.

An enhancement NMOS is voltage controlled. It is off when $V_{GS} < V_{TH}$, operates in triode/ohmic region when $V_{DS} < V_{GS} - V_{TH}$, and saturation when $V_{DS} \geq V_{GS} - V_{TH}$. A basic long-channel saturation model is

$$I_D \approx \frac{k}{2}(V_{GS} - V_{TH})^2(1 + \lambda V_{DS}). \quad (20)$$

In a switching NMOS, low $R_{DS(on)}$ minimizes I^2R loss. The insulated gate draws little DC current but has gate charge: a driver must charge/discharge it quickly. A MOSFET body diode matters in bridge and inductive-load circuits.

Feature	BJT	MOSFET
Control	Current-driven base-emitter junction	Voltage-driven insulated gate
On-state loss	$V_{CE(sat)}I_C$ in saturation	$I_D^2 R_{DS(on)}$
Input requirement	Continuous base current	Gate-charge pulses, near-zero DC gate current
Strength	Predictable analog gain, low-cost small signal	Fast, efficient switching; high input impedance
Key caution	Thermal runaway risk; forced β for switching	Gate oxide ESD; body diode and $V_{GS,max}$

Worked example: LED resistor and MOSFET switch

A 5 V supply drives a red LED with $V_F = 2$ V at 10 mA. Choose $R = (5 - 2)/0.010 = 300 \Omega$; 330Ω is a common safe value. For a low-side NMOS switching a 2 A load with $R_{DS(on)} = 50 \text{ m}\Omega$, the approximate conduction loss is $I^2R = 2^2(0.05) = 0.20$ W. Verify the data-sheet thermal resistance, voltage rating, gate-drive voltage, and inductive flyback path before selecting the device.

6 Operational Amplifiers and Analog Building Blocks

6.1 Ideal op-amp model and negative feedback

For an ideal op-amp, input current is zero, open-loop gain is infinite, and, with negative feedback in linear operation, $v_+ = v_-$. Real devices have finite gain/bandwidth, input bias and offset, limited common-mode range, finite output current, finite slew rate, and output swing limited by their supply rails.

Circuit	Connection / transfer	Main use
Voltage follower	$v_o = v_i$	Buffer; high input, low output impedance
Inverting amplifier	$v_o = -(R_f/R_{in})v_i$	Gain and inversion
Non-inverting amplifier	$v_o = (1 + R_f/R_g)v_i$	Gain without inversion
Summing amplifier	$v_o = -R_f \sum (v_k/R_k)$	Weighted addition
Difference amplifier	$v_o = (R_2/R_1)(v_2 - v_1)$ if ratios match	Subtraction / sensing
Transimpedance	$v_o = -i_{in}R_f$	Photodiode/current sensor
Comparator	output saturates high/low	Threshold decision; no linear feedback

Worked example: Non-inverting gain design

Design a non-inverting amplifier with gain 11. Since $A_v = 1 + R_f/R_g$, choose $R_g = 10 \text{ k}\Omega$ and $R_f = 100 \text{ k}\Omega$. A 0.20 V input ideally produces 2.20 V. Confirm that this output is inside the chosen op-amp's supply and output-swing limits, and that the required closed-loop bandwidth exceeds the signal frequency.

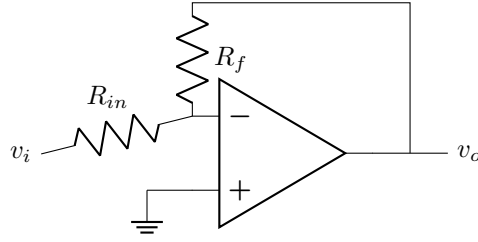


Figure 6: Inverting op-amp amplifier under negative feedback.

7 Transformers, Machines, and Introductory Power Systems

7.1 Ideal transformer and practical losses

For an ideal transformer with turns ratio $a = N_p/N_s$:

$$\frac{V_p}{V_s} = a, \quad \frac{I_p}{I_s} = \frac{1}{a}, \quad Z_{\text{seen at primary}} = a^2 Z_L, \quad P_p = P_s. \quad (21)$$

It requires changing flux, so it does not continuously transform DC. Real transformers have copper loss ($I^2 R$), core hysteresis/eddy-current loss, leakage reactance, magnetizing current, regulation, and insulation limits.

7.2 Machine fundamentals

Machine	Principle	Key relation / feature
DC machine	Commutator maintains unidirectional torque	$E \propto \Phi\omega$, $T \propto \Phi I_a$
Induction motor	Rotor current induced by rotating stator field	$n_s = 120f/P$, $s = (n_s - n_r)/n_s$
Synchronous machine	Rotor field locks to rotating stator field	$n_r = n_s = 120f/P$ in steady operation

An induction motor must run below synchronous speed in motoring operation so that slip induces rotor current. Increasing mechanical load generally increases slip and rotor current. Motor starting current can be large; protection and starting method matter.

7.3 Power-system essentials

A power system has generation, transformers, transmission, distribution, switchgear, loads, grounding, and protection. A one-line diagram represents a balanced three-phase network with one conductor. Per-unit normalizes quantities:

$$Z_{base} = \frac{V_{LL,base}^2}{S_{3\phi,base}}, \quad I_{base} = \frac{S_{3\phi,base}}{\sqrt{3}V_{LL,base}}, \quad x_{pu} = \frac{x}{x_{base}}. \quad (22)$$

Circuit breakers interrupt fault current; fuses open on overcurrent; relays sense conditions and command breakers. Grounding provides a defined reference and a fault-current path; it does not make energized conductors safe to touch.

Worked example: Transformer impedance reflection

A 120 V:24 V ideal transformer supplies a 6Ω load. The primary-to-secondary ratio is $a = 120/24 = 5$, so the primary sees $Z_{in} = a^2 Z_L = 25(6) = 150 \Omega$. The secondary current is $24/6 = 4$ A and the primary current is $4/5 = 0.8$ A; both sides carry 96 W ideally.

8 Digital Electronics and Data Conversion

8.1 Logic fundamentals

Logic signals represent 0 and 1 within guaranteed voltage ranges. Essential Boolean identities include $\overline{\overline{A}} = A$, $A + \overline{A} = 1$, $A\overline{A} = 0$, and De Morgan's laws:

$$\overline{AB} = \overline{A} + \overline{B}, \quad \overline{A + B} = \overline{A}\overline{B}. \quad (23)$$

AND, OR, and NOT are universal building blocks; NAND and NOR alone can implement any Boolean function. XOR is 1 when inputs differ. Combinational logic depends only on current inputs; sequential logic also stores state.

Element	Function	Clocked?	Typical use
Multiplexer	Select one input	No	Data routing
Decoder	Activate one output from binary code	No	Address selection
D flip-flop	Store D at active clock edge	Yes	Register / synchronizer
Counter	Advance stored binary state	Yes	Timing / frequency division
Register	Group of flip-flops	Yes	Multi-bit storage

8.2 ADC, DAC, sampling, and resolution

An N -bit ADC has 2^N codes; for input range V_{FS} , its ideal LSB size is approximately $V_{FS}/2^N$. To avoid aliasing for a band-limited signal, sample above twice its highest frequency: $f_s > 2f_{max}$, and use an anti-alias low-pass filter before the ADC. A DAC converts code to analog voltage; its reference accuracy, integral/differential nonlinearity, settling time, and output drive matter in practice.

Worked example: ADC resolution

A 12-bit ADC spans 0 V to 3.3 V. Its ideal step size is $3.3/2^{12} = 0.806$ mV per count. The quantization uncertainty is commonly modeled as about ± 0.5 LSB, before reference, offset, gain, noise, and nonlinearity errors are included.

9 Feedback Control Fundamentals

9.1 Transfer functions, poles, and feedback

For an LTI system, $G(s) = Y(s)/U(s)$ under zero initial conditions. A pole is a root of the denominator and a zero is a root of the numerator. With negative feedback $H(s)$, the closed-loop transfer function is

$$T(s) = \frac{G(s)}{1 + G(s)H(s)}. \quad (24)$$

Negative feedback can reduce sensitivity and improve disturbance rejection, but excess phase lag can make the loop oscillate. A stable continuous-time system has all closed-loop poles in the left half of the s plane.

9.2 First-order response and PID intuition

A first-order plant $G(s) = K/(\tau s + 1)$ has time response $y(t) = K(1 - e^{-t/\tau})u(t)$ for a unit step. The 10–90% rise time is about 2.2τ . Proportional control responds to current error, integral control removes constant steady-state error, and derivative action anticipates changing error but amplifies noise:

$$C(s) = K_p + \frac{K_i}{s} + K_d s. \quad (25)$$

Gain margin and phase margin measure distance from instability in frequency response. Saturation requires anti-windup when integral action is used.

10 Measurement, Grounding, and Electrical Safety

10.1 Making trustworthy measurements

Instrument	Strength	Main limitation / check
Digital multimeter	Accurate DC/low-frequency voltage, current, resistance	Burden voltage in current mode; input impedance
Oscilloscope	Time-domain waveform, transients, noise	Probe rating, bandwidth, sampling, earth-referenced ground clip
Current probe	Nonintrusive current waveform	Range, bandwidth, offset/degauss
LCR meter	Component value and loss at stated frequency	Test frequency and fixture parasitics

Bandwidth must exceed the signal content; a rule of thumb for a single-pole instrument is $t_r \approx 0.35/BW$. Use a $10\times$ oscilloscope probe when its reduced loading and higher voltage rating are appropriate, and compensate it before accurate fast-edge measurement. Report uncertainty from instrument specification, calibration, repeatability, and setup effects rather than only display digits.

10.2 Grounding and safety

Protective earth (PE) bonds exposed conductive enclosures to reduce touch-voltage hazard and encourage fault current to clear protection. Signal ground is a circuit reference; it may be tied to earth at a deliberate point. Avoid accidental ground loops and never clip an earth-referenced oscilloscope ground lead to a floating high-side node unless the measurement arrangement is designed for it. Use a differential probe or isolated measurement system when required.

Caution: Safe work is a design requirement

De-energize before changing wiring when possible. Apply lockout/tagout, verify absence of voltage with a rated meter, use equipment with adequate voltage and fault-category ratings, and inspect leads, fuses, and insulation. Treat stored energy in capacitors, rotating equipment, batteries, and inductors as hazardous. Follow the applicable site procedure and local electrical-safety standard; this guide is not a substitute for them.

11 Formula Review

DC: $V = IR$; $P = VI = I^2R = V^2/R$; $R_s = \sum R$; $1/R_p = \sum 1/R$; $V_o = V_s R_2/(R_1 + R_2)$; $R_L = R_{th}$ for maximum resistive power.

Energy and transients: $w_C = CV^2/2$; $w_L = LI^2/2$; $i_C = Cdv/dt$; $v_L = Ldi/dt$; $\tau_{RC} = RC$; $\tau_{RL} = L/R$; $x = x_\infty + (x_0 - x_\infty)e^{-t/\tau}$.

AC: $V_{rms} = V_m/\sqrt{2}$; $Z_R = R$; $Z_L = j\omega L$; $Z_C = 1/(j\omega C)$; $S = VI^* = P + jQ$; $P = VI \cos \phi$; $Q = VI \sin \phi$; $\text{pf} = \cos \phi$.

Filters and resonance: $H_{LP} = 1/(1+sRC)$; $H_{HP} = sRC/(1+sRC)$; $f_c = 1/(2\pi RC)$; $\omega_0 = 1/\sqrt{LC}$.

Three phase: $S_{3\phi} = \sqrt{3}V_{LL}I_L^*$; $P = \sqrt{3}V_{LL}I_L \cos \phi$; $Y: V_L = \sqrt{3}V_\phi$, $I_L = I_\phi$; $\Delta: V_L = V_\phi$, $I_L = \sqrt{3}I_\phi$.

Devices: $I_C \approx \beta I_B$; $I_D \approx k(V_{GS} - V_{TH})^2/2$; $A_{inv} = -R_f/R_{in}$; $A_{noninv} = 1 + R_f/R_g$.

Transformer and machines: $V_p/V_s = N_p/N_s$; $Z_p = a^2 Z_s$; $n_s = 120f/P$; $s = (n_s - n_r)/n_s$.

Digital and control: $\text{LSB} \approx V_{FS}/2^N$; $f_s > 2f_{max}$; $T = G/(1+GH)$; $C_{PID} = K_p + K_i/s + K_d s$.

12 Mixed Practice Problems with Answers

1. A 24 V source drives series 3Ω and 9Ω resistors. Find current, voltage across 9Ω , and its power.
2. An initially charged $10\mu\text{F}$ capacitor at 5 V discharges through $100\text{k}\Omega$. Write $v_C(t)$ and find v_C at one time constant.
3. A 100V_{rms} supply feeds a 25Ω resistor. Find average power and peak current.
4. A balanced Y load has $Z_\phi = 8 + j6\Omega$ and is supplied by 208 V line-to-line. Find I_L magnitude and total real power.
5. Find the inverting-op-amp output for $R_{in} = 5\text{k}\Omega$, $R_f = 50\text{k}\Omega$, and $v_i = 0.30\text{V}$.

6. A four-pole induction motor operates from 60 Hz and runs at 1740 rpm. Find synchronous speed and slip.

Answers

1. $I = 24/(3 + 9) = 2 \text{ A}$; $V_{9\Omega} = 18 \text{ V}$; $P_{9\Omega} = 18(2) = 36 \text{ W}$.
2. $\tau = RC = (100\,000)(10 \times 10^{-6}) = 1 \text{ s}$ and $v_C(t) = 5e^{-t} \text{ V}$; at $t = \tau$, $v_C = 1.84 \text{ V}$.
3. $P = V^2/R = 100^2/25 = 400 \text{ W}$; $I_{rms} = 4 \text{ A}$ and $I_{peak} = 4\sqrt{2} = 5.66 \text{ A}$.
4. $|Z| = 10 \Omega$, $V_\phi = 208/\sqrt{3} = 120 \text{ V}$, $I_L = I_\phi = 120/10 = 12 \text{ A}$. With $\text{pf} = 8/10 = 0.8$, $P = \sqrt{3}(208)(12)(0.8) = 3.46 \text{ kW}$.
5. $v_o = -(50/5)(0.30) = -3.0 \text{ V}$, provided the supply rails permit it.
6. $n_s = 120(60)/4 = 1800 \text{ rpm}$ and $s = (1800 - 1740)/1800 = 0.0333$, or 3.33%.